PUBUDU GUNASEKARA

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MS Computer Science, Northeastern University | Seeking Software Engineering Internship, Summer 2027

EDUCATION

Northeastern University, Khoury College of Computer Sciences, San Jose, CA Jan 2026 – May 2028 (Expected)
M.S. in Computer Science | GPA: 4.0 / 4.0
Coursework: Algorithms, Programming Design Paradigms; in progress: Machine Learning, NLP (RAG system)

Conestoga College, Postgraduate Certificate in Mobile Solutions Development, GPA 3.74 / 4.0 2022 – 2023

Victoria University, Bachelor of Information Technology, GPA 6.25 / 7.0 2018 – 2021

EXPERIENCE

Food Services & Hospitality, Waterloo, Canada Jan 2024 – Dec 2025

Virtusa, Associate Engineer (QA), Colombo, Sri Lanka Jun 2021 – Mar 2022

- Wrote **Java and Selenium WebDriver** automated tests inside the team's framework for a British Telecommunications (BT) platform serving **1M+ users**. Designed and ran test plans across agile sprints.
- Finished assigned sprint tickets ahead of the sprint end, leaving room for peer review and ad hoc requests. Ran client demos and daily stand-ups when the QA lead delegated them, and tracked defect density and open defect counts.

PROJECTS

Distributed Task Scheduler Code · Case Study
Java 21, Spring Boot, Kafka, Redis, PostgreSQL, Prometheus, Docker, GitHub Actions

- Built a task scheduler on **hexagonal architecture** (enforced with ArchUnit) that guards against duplicate execution under Kafka's at-least-once delivery, using a database state machine, optimistic locking and a Redis distributed lock.
- Handles failures with 3 retries (10s, 30s, 90s backoff), a dead-letter topic, and a recovery job that re-queues stuck tasks and re-publishes lost creation events. Redis rate limiting protects the API.
- **188 tests** with **93.4% line and 91.1% branch coverage** (80% gate in CI); integration tests run against real PostgreSQL and Redis through Testcontainers.

AI Code Review Assistant Code · Live Demo
React 19/Vite, Node.js/Express, PostgreSQL, Redis, gpt-4o-mini, Docker, AWS

- Shipped a GitHub OAuth app (React on AWS Amplify, Express on EC2) that reviews a pull request and returns findings tagged by severity and category, each with a CWE reference and suggested fix.
- Strict JSON-schema output (OpenAI Structured Outputs) that fails a review rather than inventing results, a per-user hourly rate limit (atomic Lua script) and a Redis cache keyed on diff, model and prompt version. 19 backend tests.

Smart Farm Code | Demo
React Native (Expo), Redux, Firebase, Arduino, ESP

- Built the app's **device-control section** and the physical Arduino farm model. The control panel (water, food, gate, cleaning) sends commands through Firebase to WiFi-enabled Arduino/ESP controllers that drive relays and motors.
- Team project, one of the **Top 10 Most Innovative Projects** at the NSBM Green EXE v1.0 software competition.

Travel Day Scheduler Code
Python, Weighted A Search*

- Graduate algorithms team project (team of 2): a weighted / anytime A* itinerary planner with heap pruning. Wrote the initial search and the route-visualization frontend.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL, C++ (Arduino)
Backend & Distributed: Spring Boot, Node.js / Express, Apache Kafka, Redis, PostgreSQL, Firebase, REST APIs
Infra & DevOps: Docker, Docker Compose, GitHub Actions, CI/CD, Jenkins, Prometheus, Grafana, AWS
Frontend: React, Next.js, React Native, Vite, Redux
Certifications: Microsoft Azure Fundamentals (AZ-900), IBM Full Stack Software Developer

LEADERSHIP

- Founding Cohort Member, Graduate Student Career Advisory Board, Northeastern University (Khoury College)